

Weather and Climate Big Data

기상기후빅데이터

Hands-on Exercise 2: ECMWF Reanalysis

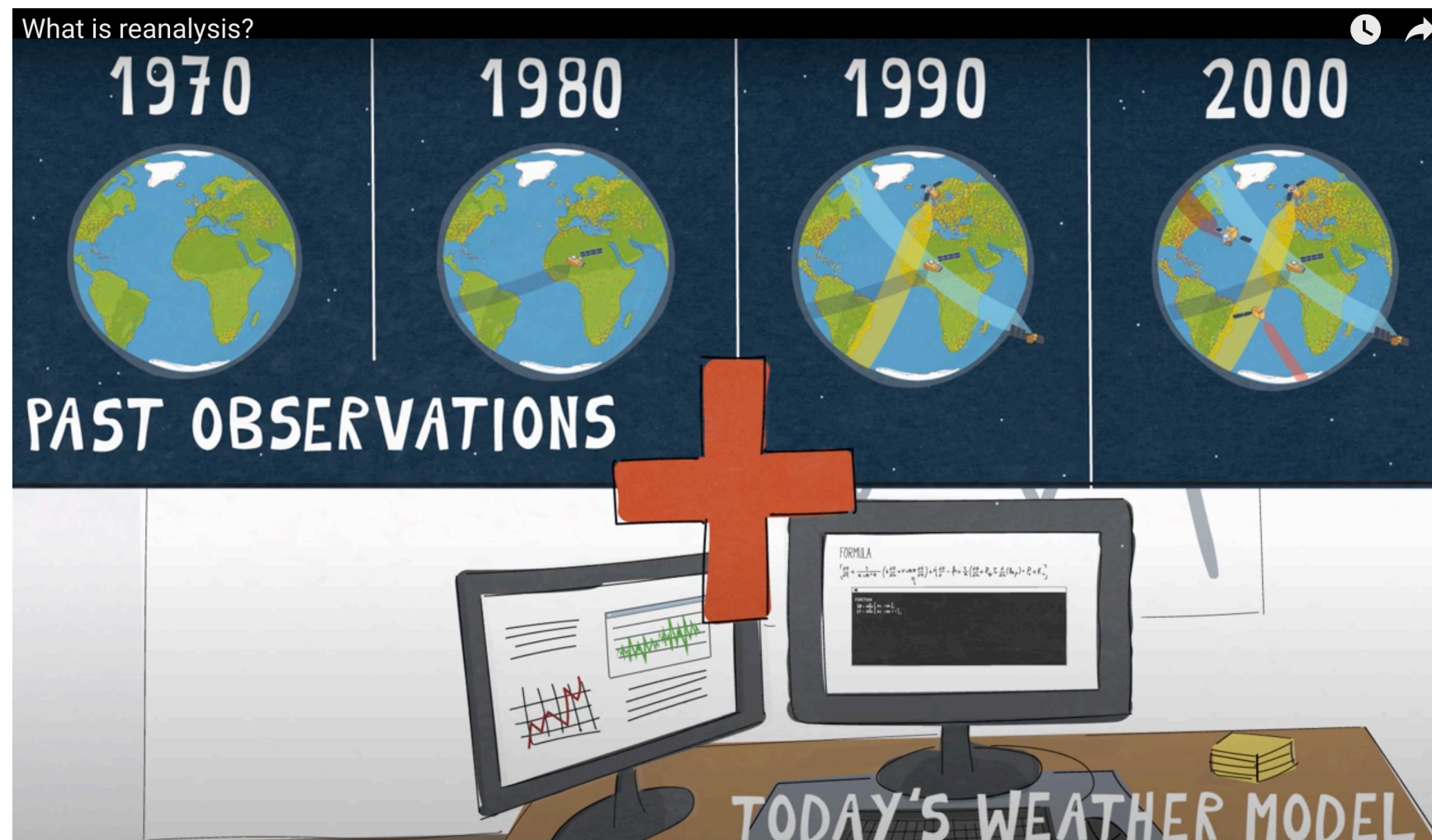
오승민
Spring 2025

Today's Topics

- Reanalysis and Data Assimilation
- Multidimensional data in NetCDF format
- Application of reanalysis data

Reanalysis

Reanalysis data refers to a comprehensive dataset that combines historical observational data with numerical weather prediction models to create a consistent representation of the Earth system over time.



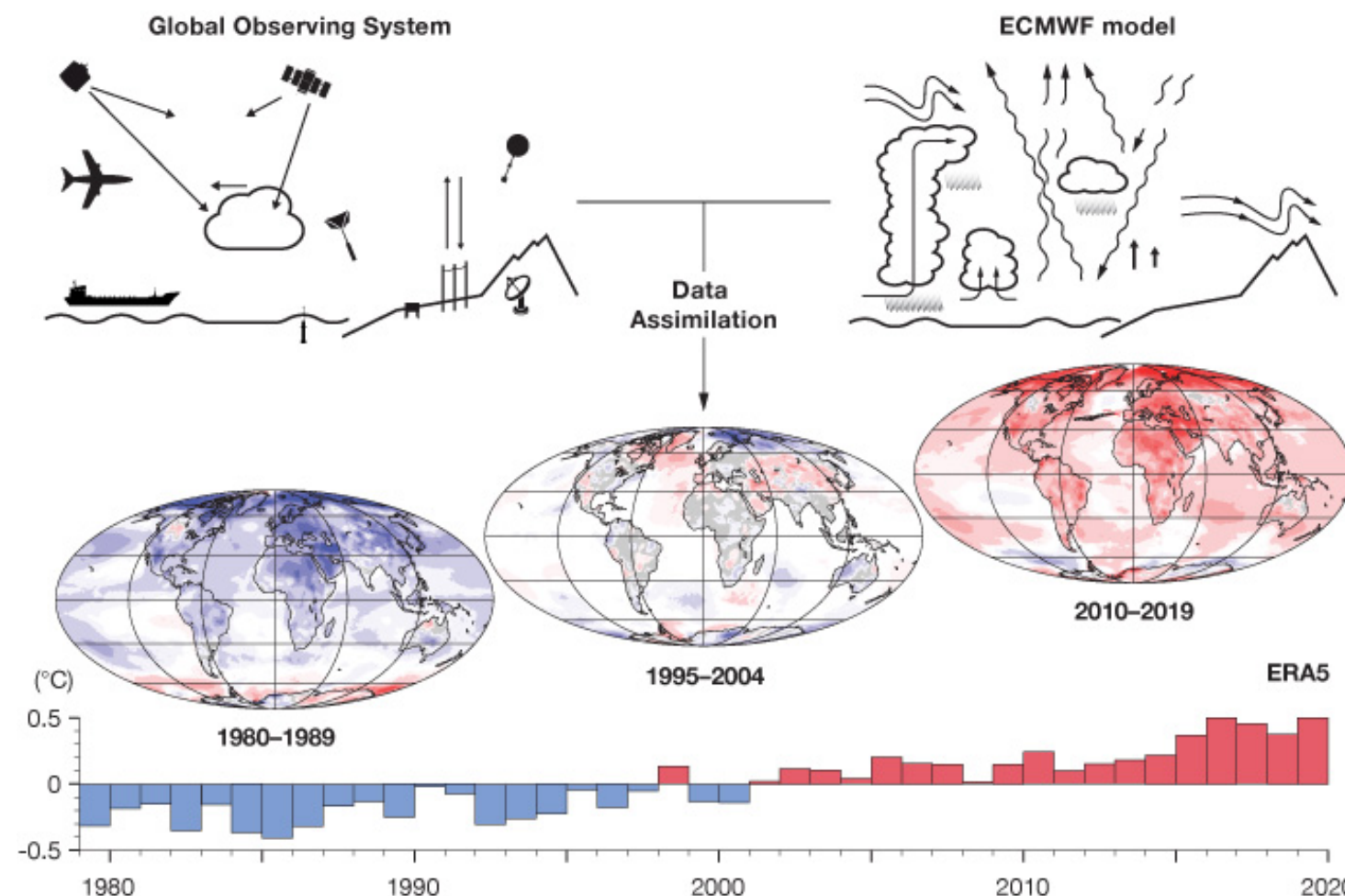
<https://www.youtube.com/watch?v=FAGobvUGl24>

Reanalysis

- **Reanalysis data:** Combines past observational data (satellites, weather stations, buoys, etc.) with modern numerical weather models to create a consistent long-term dataset.
 - Provides **spatially and temporally continuous** data on a **uniform grid**, filling gaps where direct observations are missing.
 - Uses real observations to **correct biases and keep model outputs closer to actual conditions**, improving long-term accuracy.

Data Assimilation

In reanalysis, observations from weather stations, aircraft, ships, satellites and other sources are combined with a state-of-the-art weather model in a process called *data assimilation*.

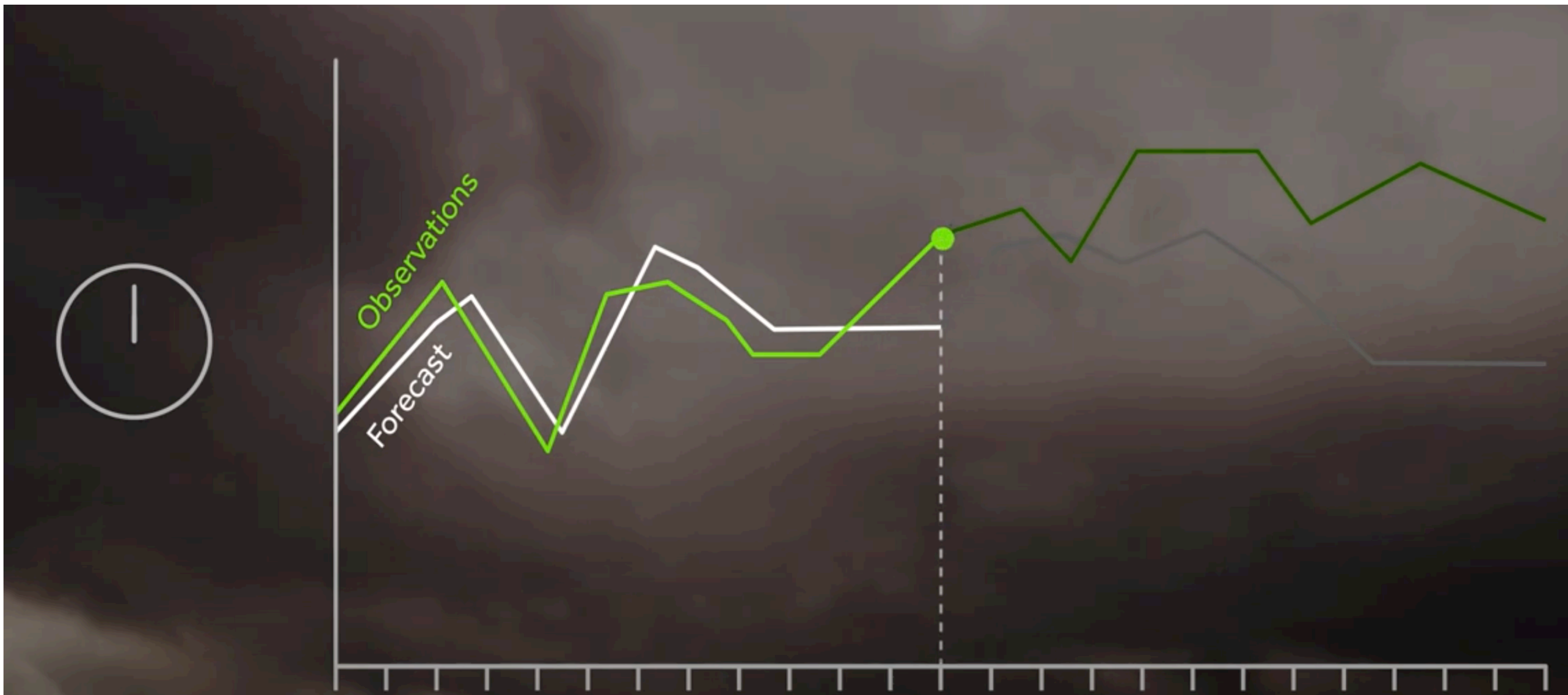


Data Assimilation

Initial Condition Problem



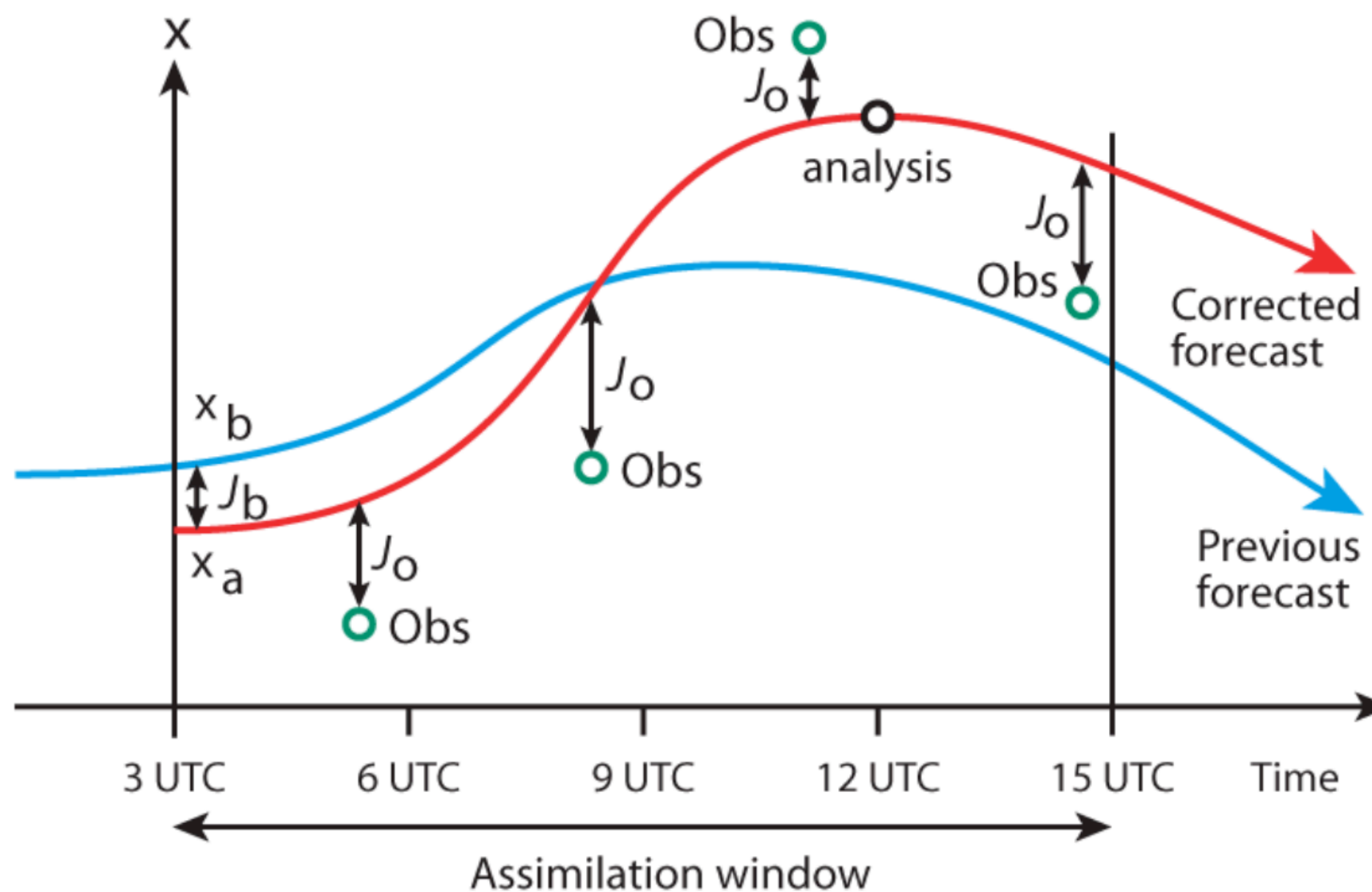
Data Assimilation



Data Assimilation



Data Assimilation



Examples

- Modern-Era Retrospective analysis for Research and Applications Version 2 (MERRA-2), produced by NASA
- JRA-55 produced by the Japan Meteorological Agency (JMA)
- Climate Forecast System Reanalysis (CSFR) produced by the National Centers for Environmental Prediction (NCEP)
- Twentieth Century Reanalysis (20CR) produced by the National Oceanic and Atmospheric Administration (NOAA)
- ECMWF Reanalysis (ERA5), produced by ECMWF

Examples: ERA5



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Climate Data Store

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User guide

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28 Sep 2024

Welcome to the New Climate Data Store (CDS)! This new system is in its early days of full operations and still undergoing enhancements and fine tuning. Some disruptions are to be expected. Your [feedback](#) is key to improve the user experience on the new CDS for the benefit of everyone. Thank you.

< 1/5 >

Search results

era5



Search

Atmosphere (surface) ×

Atmosphere (upper air) ×

Atmosphere (upper level) ×

Ocean (physics) ×

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Filter by

Product type



Provider



Sector



Spatial coverage

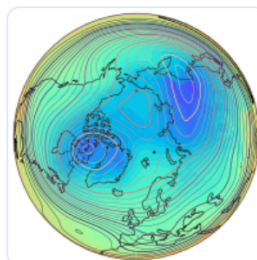


Temporal coverage



We found 31 results

Sort by: Relevance



⚠ The final validated ERA5 differs from ERA5T in July 2024 - please refer to our [Forum announcement](#) for details and watch it for further updates on this.

ERA5 monthly averaged data on pressure levels from 1940 to present

Updated on 8 October 2024

ERA5 is the fifth generation ECMWF reanalysis for the global climate and weather for the past 8 decades. Data is available from 1940 onwards. ERA5 replaces the ERA-Interim reanalysis. Reanalysis combines model data with observations from across the world into a globally complete and consistent dataset...

Reanalysis

Copernicus C3S

Global

Past

Atmosphere (surface)

Atmosphere (upper air)



Examples: ERA5

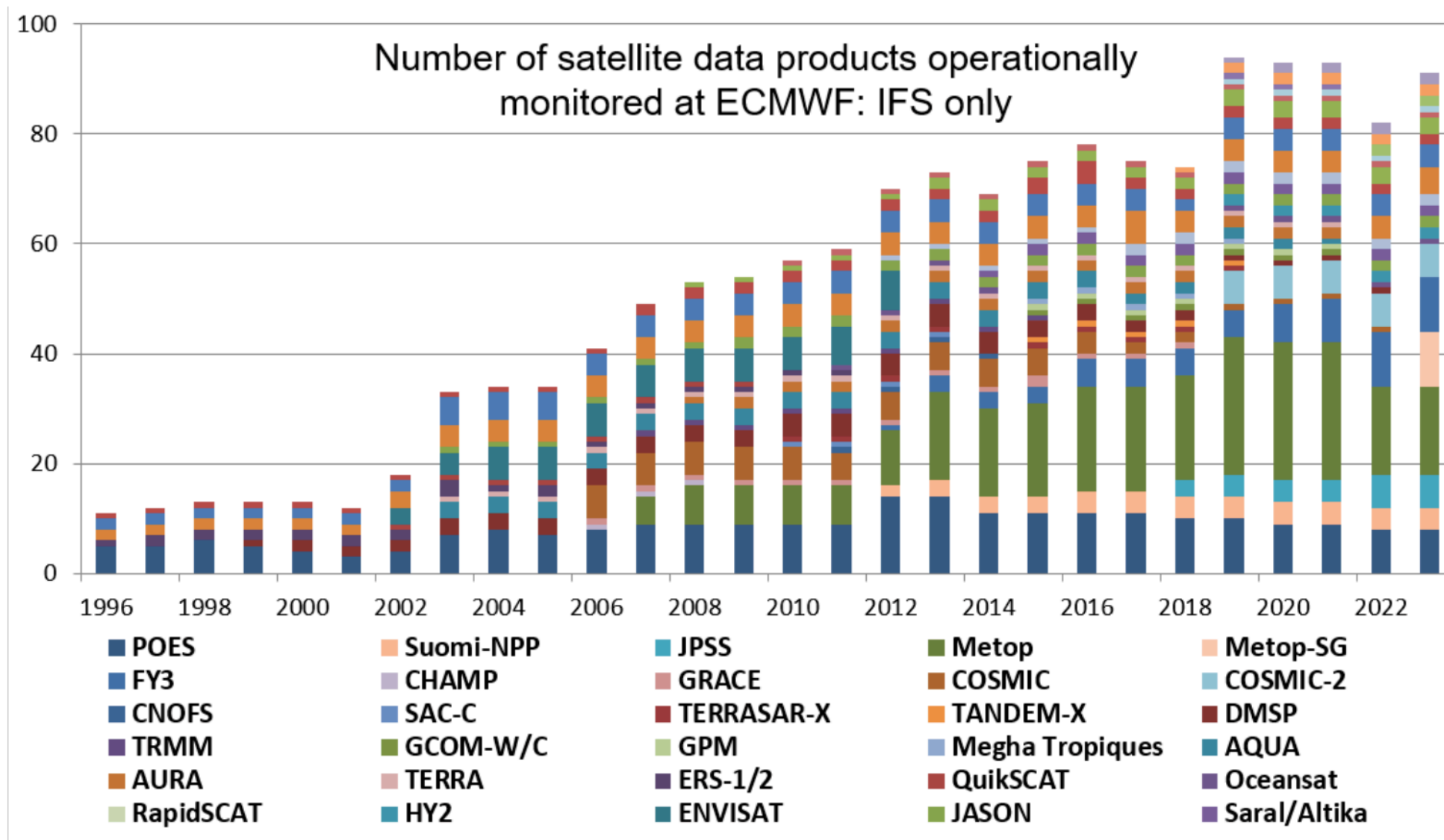
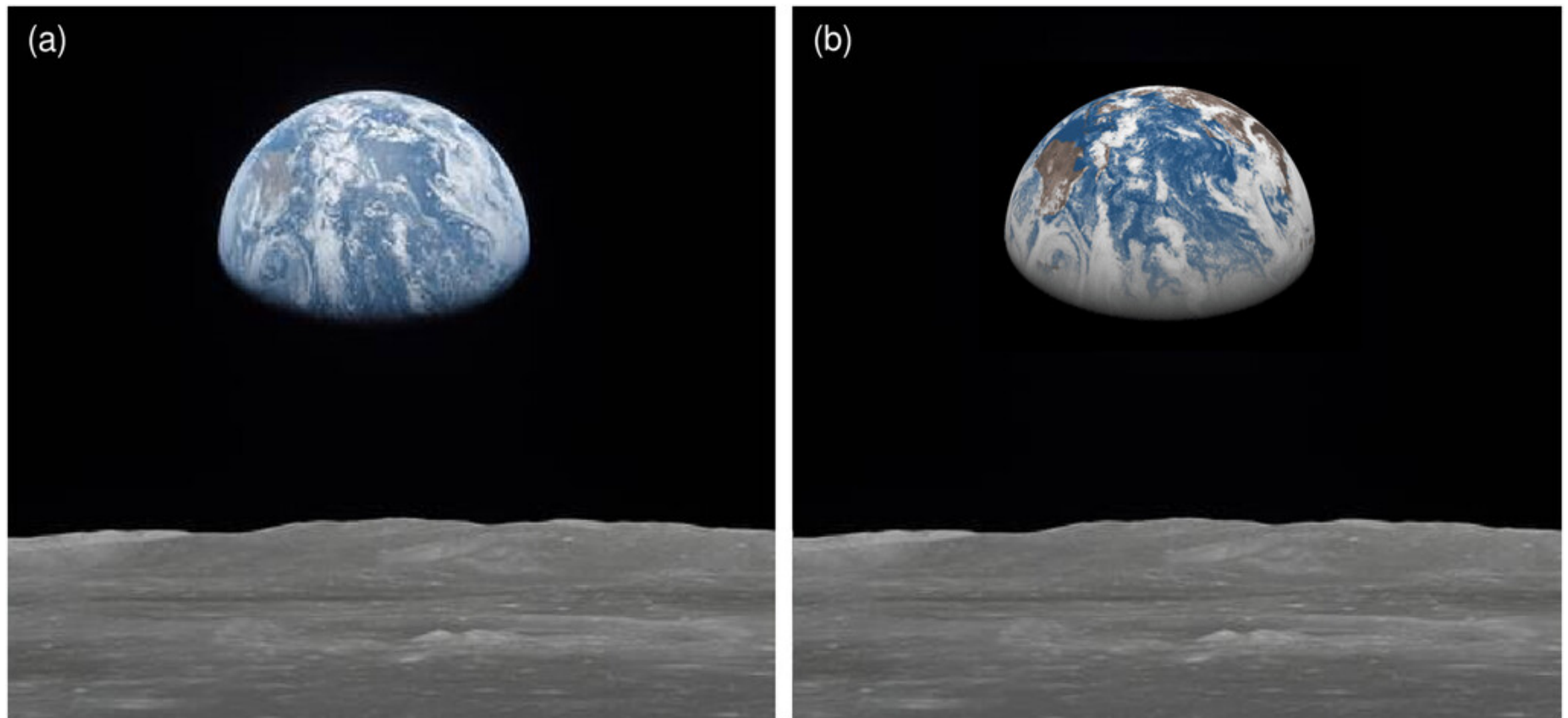


Figure 3. Evolution of the number of satellite instruments processed (or expected to be processed) in the ECMWF Integrated Forecasting System (IFS) during the period 1996 - 2023.

Examples: ERA5

Cloud patterns on earth observed by Apollo 11 on 20 July 1969 versus reanalysis 2020



Data Download



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 Sungmin O

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26 Sep 2024 Watch our [Forum](#) for Announcements, news and other discussed topics.

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Dive into this wealth of information about the Earth's
past, present and future climate

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Data Download

★ ERA5 post-processed daily statistics on single levels from 1940 to present

- Overview **Download** Documentation

Complete all required fields before submitting the request. [Clear all fields](#)

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References
[Citation and attribution](#)
DOI: [10.24381/cds.4991cf48](#)

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Publication date
2024-10-14 

<https://cds.climate.copernicus.eu/datasets/derived-era5-single-levels-daily-statistics?tab=overview>

Data Download

API Token

Data download requests can be sent programmatically via [our API](#). However, an API token must be provided.

API Token

.....

[👁](#) [🔄](#) [📄](#)

Details

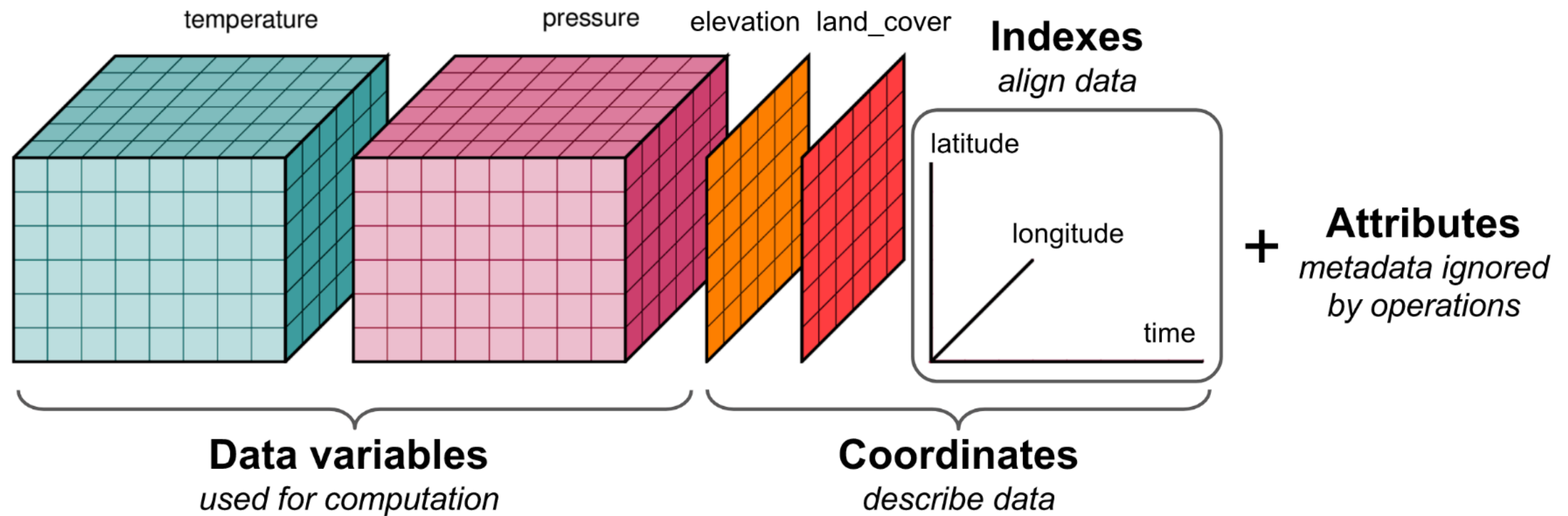
Country	Korea, Republic of	✎
Affiliation	Research and education organisations	✎
Organisation (Optional)	Edit	✎
Thematic activities	Climate change	✎
Activity sectors	Research and education	✎



Network Common Data Form (NetCDF)

- **NetCDF is both a file format and a software library** used for storing, accessing, and sharing scientific array-oriented data (e.g. climate, weather, and ocean data).
- **Self-describing & machine-independent:** NetCDF files contain metadata (variables, dimensions, attributes) and can be used across different platforms without compatibility issues.
- **Supports multiple programming languages** (Python, C, Fortran, Java) and provides efficient data compression and access.

NetCDF



NetCDF

◆ netCDF4

python

📄 복사

✎ 편집

```
from netCDF4 import Dataset
f = Dataset('file.nc')
temp = f.variables['t2m'][0, :, :]
f.close()
```

◆ xarray

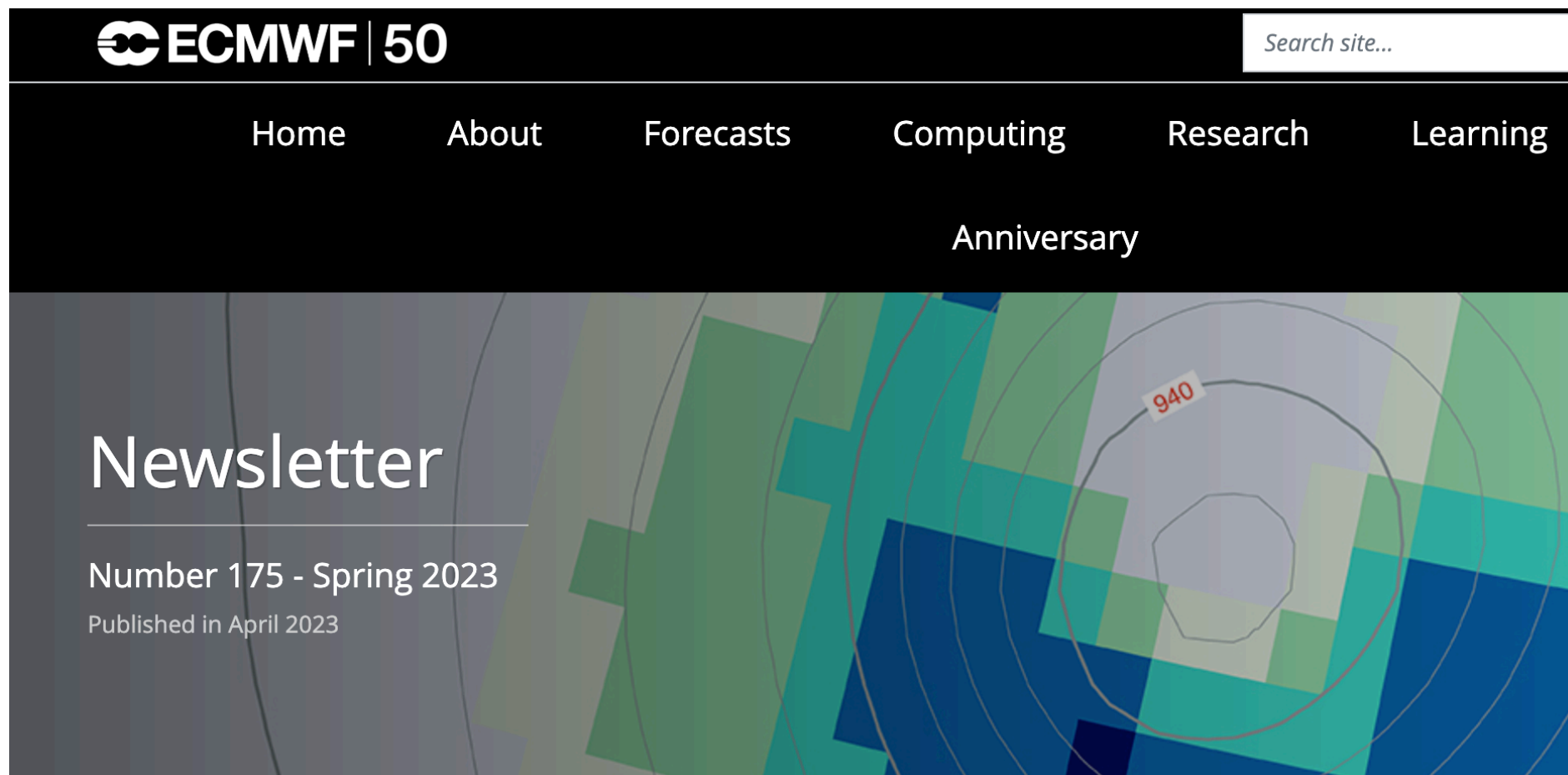
python

📄 복사

✎ 편집

```
import xarray as xr
ds = xr.open_dataset('file.nc')
temp = ds['t2m'].isel(time=0)
```


Use of Reanalysis



NEWS ERA5 reanalysis now available from 1940

[◀ View all Newsletters](#)

Hans Hersbach

Editorial

Use of Reanalysis

GraphCast for weather forecasting

A re-implementation of the DeepMind's GraphCast model in PhysicsNeMo.

Problem overview

GraphCast is a multi-scale graph neural network-based autoregressive model. It is trained on historical weather data from ECMWF's ERA5 reanalysis archive. GraphCast generates predictions at 6-hour time steps for a set of surface and atmospheric variables. This prediction covers a 0.25-degree latitude-longitude grid, providing approximately 25 x 25 kilometer resolution at the equator.

Dataset

The model is trained on a 73-channel subset of the ERA5 reanalysis data on single levels and pressure levels that are pre-processed and stored into HDF5 files. Additional static channels such as land-sea mask and geopotential, and the cosine zenith angle are also used. The ERA5 dataset can be downloaded here. A curated 20-channel subset of the ERA5 training data in HDF5 format is hosted at the National Energy Research Scientific Computing Center (NERSC). For convenience it is available to all via Globus. You will need a Globus account and will need to be logged in to your account in order to access the data. You may also need the Globus Connect to transfer data.

Use of Reanalysis

Why now? 3 key factors

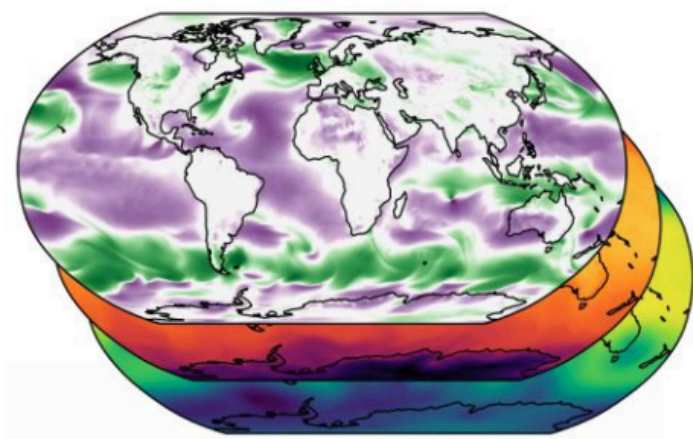
Data

ERA5 reanalysis

Assimilation is still NWP-based!

Massive dataset (40+ years)

High-quality data



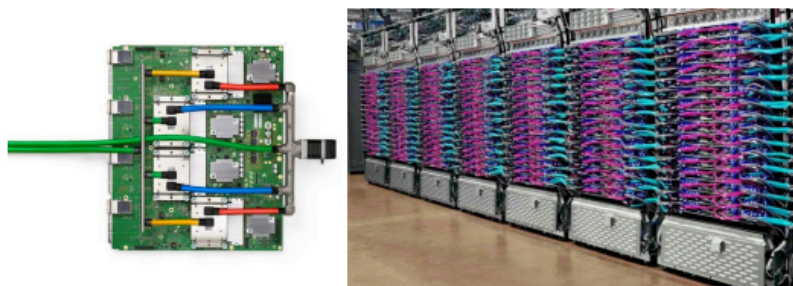
Compute

32 TPUs

Increase in scale in ML models

Parallel compute

1 TPU-minute vs 10k CPU



Deep Learning algorithms

Graph Neural Networks

Encoded inductive biases

Locality

Equivariance

